

Two ways to watch a room clear

Two independent simulation methods, pointed at the same room, disagreed about how much benefit its UV fixtures actually provide — by nearly a factor of two, consistently. This is the log of tracking that disagreement to its source, one hypothesis at a time.

Room patient ward, 4 fixtures **Method A** transient decay (start contaminated, watch it clear)
Method B continuous-source steady state (constant leak, watch it equilibrate)
Combinations examined 6 – three susceptibility levels × two ventilation rates

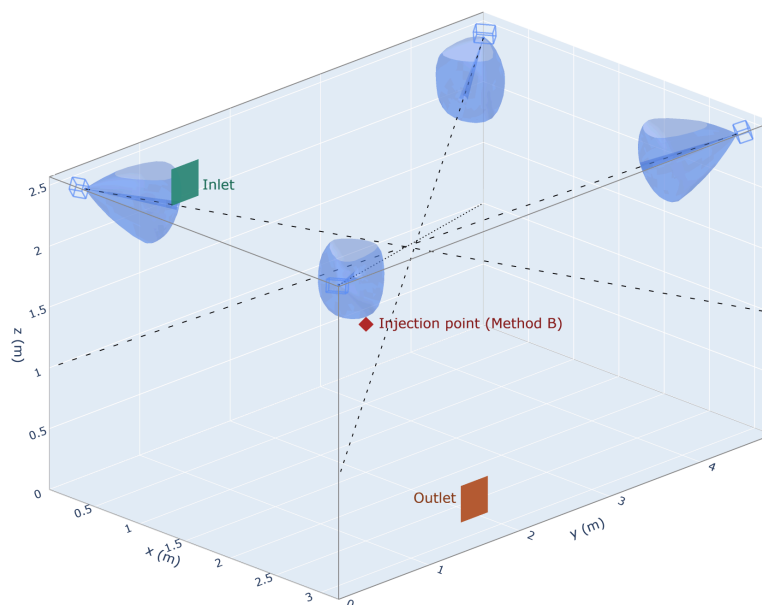
Background

What's actually being simulated

Germicidal UV (GUV) fixtures disinfect room air in place, alongside whatever mechanical ventilation the room already has. To compare the two side by side, UV's contribution gets expressed in the same currency as ventilation: *equivalent air changes per hour* ($eACH_{uv}$) — the ventilation rate that would remove contaminant just as fast as the UV fixtures actually do. A room with 6 ACH of ventilation and an $eACH_{uv}$ of 35/hr is, for disinfection purposes, behaving like a room with ~41 air changes per hour.

The room itself is a small patient ward — 3.2 m × 4.8 m × 2.57 m — fitted with four upper-room UV fixtures aimed across the space above head height. The CFD model (OpenFOAM) resolves the room's actual airflow at roughly 40,000 mesh cells: inlet, outlet, the fixtures' UV dose field, and a passive contaminant field T that gets carried around by that airflow and removed by both ventilation and UV.

- Inlet (supply air)
- Outlet (exhaust air)
- ◆ Contaminant injection (Method B)



Units: meters

The room as modeled: $3.2 \times 4.8 \times 2.57$ m, four upper-room UV fixtures (cones show each fixture's emission pattern, dashed lines its aim direction), plus the inlet, outlet, and Method B's contaminant injection point. Mesh, airflow, and dose field are all resolved against this same geometry by both methods below.

Getting $eACH_{uv}$ out of a CFD room requires actually simulating contamination, not just UV dose — and there are two legitimate, physically different ways to do that:

Method A — transient decay

Start the room uniformly contaminated (as if freshly mixed), turn on ventilation and UV together, and watch the concentration fall over real simulated time. Fit an exponential to the decay curve; its rate constant, converted to the same units as ACH, is the total removal rate — ventilation and UV combined. A second, UV-off control run (identical room, no UV) isolates ventilation's own share, and subtracting it leaves $eACH_{uv}$.

Method B — continuous-source steady state

Instead of a decay curve, leak contaminant continuously from a small point at a known, fixed rate and let the room find its own equilibrium concentration — first

under ventilation alone (Phase 1), then with UV added (Phase 2). At equilibrium, injection and removal are equal by definition, so the room's average concentration directly implies the total removal rate: a lower steady-state concentration for the same fixed leak means removal is working harder. Comparing Phase 1's and Phase 2's equilibrium concentrations isolates UV's share the same way the control run does for Method A.

Both methods are standard, defensible ways to get the same physical quantity out of a CFD model. Run against the identical room, fixtures, and ventilation setting, they should agree.

Entry 1

OBSERVED

The same room, two different answers

Both methods report the same headline number: *eACH_{uv}*, the equivalent air changes per hour a room's UV fixtures are worth — a way of expressing "how much disinfection" in the same units as ventilation, so the two can be compared directly. Run the identical room, identical fixtures, identical ventilation setting through both methods, and the physics is the same system either way. The numbers should converge.

They didn't.

EACH_UV (/HR), SIX MATCHING COMBINATIONS

Z	ACH	METHOD A (DECAY)	METHOD B (STEADY-STATE)	RATIO
1.7	6	10.96	18.99	1.73×
1.7	9	11.18	19.59	1.75×
6	6	34.85	63.44	1.82×
6	9	36.50	64.08	1.76×
8.5	6	46.79	91.02	1.95×
8.5	9	49.78	89.97	1.81×

Not scattered, not noise — a ratio clustered tightly between 1.73× and 1.95× across a 5× range of susceptibility and two different ventilation rates. Whatever was driving it had to be systematic, not incidental.

Entry 2

CONFIRMED – NOT THE MAIN CAUSE

Hypothesis: we weren't measuring "ventilation" the same way

The air-change rate you dial in is a boundary condition, not a guarantee — real rooms short-circuit, and the air that actually reaches the bulk of the room can move quite differently from what the inlet nominally delivers. Both methods try to measure the room's *actual* ventilation rate rather than trust the dial. They did it differently: Method A runs a dedicated experiment — pre-mix the room, add no source, watch the concentration clear on ventilation alone. Method B, as originally built, inferred the same number indirectly from a slow buildup near a small injection point.

Giving Method B the same kind of dedicated, pre-mixed experiment Method A already used produced a clean, independent check:

MEASURED VENTILATION RATE, Z=6 / ACH=6		
1.640 /hr	1.636 /hr	0.2%
Method A's own measurement	Method B, same style of experiment	difference between them

A real, independently-confirmed improvement to how Method B measures ventilation. But feeding that corrected number into Method B's disinfection estimate didn't close the gap with Method A — it opened it slightly further, in every one of the six combinations (1.73–1.95× widened to 1.81–1.97×). Ventilation measurement, once the obvious suspect, was cleared: real bug, wrong trail.

Entry 3

Hypothesis: uneven light meets uneven air

UV intensity in a real room is nowhere near uniform — measured across roughly 40,000 points in this room, it ranged from 0.35 $\mu\text{W}/\text{cm}^2$ in the dimmest corner to 429 $\mu\text{W}/\text{cm}^2$ right beside a fixture, well over a thousand-fold spread. Method B's shortcut for turning a room-average concentration into a disinfection rate implicitly assumes the strength of UV removal and the local concentration aren't systematically related in space. If the brightest zones happened to coincide with the least (or most) contaminated air, that assumption would quietly break, and the room average would stop meaning what the formula needs it to mean.

The clean way to test this: make UV strength perfectly uniform everywhere — identical in every part of the room — which makes that assumption exactly true by construction, regardless of how uneven the concentration field still is. If spatial mismatch between light and contaminant were the driver, flattening the light should pull the answer toward Method A's.

EACH_UV (/HR), Z=6 / ACH=6

34.85

Method A (the target)

64.77

Method B, real (uneven) light field

69.82

Method B, light forced perfectly uniform

It moved — the wrong way, and only slightly. Flattening the light field pushed Method B's answer further from Method A's, not closer. Whatever separates the two methods, it isn't where the light happens to be strong relative to where the contaminant happens to be.

Entry 3a

A quick side-check: is the light map too coarse?

Simulating every one of 40,000 cells' exact UV intensity individually is impractical, so the real, continuous light field gets grouped into a couple dozen

representative brightness bands before it's simulated. With a distribution this skewed — a thousand-fold range, a long thin tail of very bright cells next to a wide bulk of dim ones — it was worth checking whether that simplification was quietly smoothing away something that mattered.

ROOM-WIDE AVERAGE UV INTENSITY		
61.98 /hr	62.03 /hr	0.07%
from the true, unsimplified light field	from the simplified, binned version actually simulated	difference

Essentially exact. Not the cause — closed in minutes, back to the main thread.

Entry 4

REJECTED

Hypothesis: it's not where the light is — it's where the contamination starts

Rule out the light field, in every form tested, and one structural difference between the two methods still stands: how the contaminant enters the room in the first place. Method A starts from a room that's already thoroughly, uniformly mixed — every test it runs, including the ventilation check in Entry 2, begins from that same clean slate. Method B instead leaks contaminant continuously from a single small point and waits for the whole room to find its own equilibrium around it — a fundamentally more localized starting condition, held for the entire simulation, not just at t=0.

Repeated the comparison with that injection point spread across a much larger region of the room — roughly 18× the volume, nowhere near as point-like — to see whether a less concentrated source would bring Method B's answer toward Method A's.

EACH_UV_STEADY_STATE_CORRECTED (/HR), Z=6 / ACH=6		
64.77	64.79	0.03%
small source (original, 0.012 m³)	large source (18× the volume, 0.216 m³)	difference

No effect at all. An 18-fold change in how concentrated the injection is moved the answer by three hundredths of a percent. Whatever separates the two methods, it isn't the shape or size of the source either.

REJECTED

Entry 5

Hypothesis: Method B's equilibrium reading isn't actually settled

One structural difference hadn't been examined yet: how each method knows it's done. Method A's curve fit uses the whole decay history, so a still-moving tail feeds straight into the fit. Method B instead reads off a single number — the room's average concentration once Phase 1 or Phase 2 looks like it's stopped changing — and that "looks like it's stopped changing" check had already been caught giving a false answer once before, earlier in this same project: Phase 1's readiness check once certified a curve as flat when it was still visibly climbing, and needed a proper curve-fit extrapolation to the true $n \rightarrow \infty$ value (rather than just averaging a window of samples) to catch it. Phase 1 was fixed with a hard gate requiring that extrapolation to stabilize before its result is trusted. Phase 2 — the phase that actually produces the number Method B's corrected eACH_uv is built from — was never given the same hard gate, only the cheaper check already shown to be foolable. If Phase 2's reading were caught mid-climb the same way Phase 1's once was, reading it too early would overstate the true removal rate — the right direction to explain Method B sitting above Method A.

Rather than spend another multi-hour run testing it blind, the same curve-fit extrapolation Phase 1 already computes turned out to be sitting unused in Phase 2's own results, on disk, from runs already completed for Entries 3 and 4. Comparing Phase 2's plain windowed average against its own fitted $n \rightarrow \infty$ extrapolation directly answers the question, no new simulation required.

PHASE 2 ROOM-AVERAGE CONCENTRATION: WINDOWED READING VS.
CURVE-FIT EXTRAPOLATION TO $N \rightarrow \infty$

RUN	WINDOWED AVERAGE	EXTRAPOLATED (TRUE) VALUE	GAP
Real light field	0.027071	0.027099	0.1%
18× source volume	0.027250	0.027097	0.6%

For comparison, Phase 1 — the phase already known to need the extrapolation fix — showed an 8.9% gap between its windowed reading (0.551) and its true extrapolated value (0.605) in this same set of runs. Phase 2 shows nothing close to that: 0.1–0.6%, an order of magnitude smaller, and nowhere near large enough to explain a gap that would require Phase 2's reading to be off by roughly 45%. Phase 2 turns out to have settled well within its iteration budget after all — this just hadn't been checked directly before. Closed: a real and reasonable concern, tested against real data, and cleared.

Five hypotheses in now — one confirmed as a real, separate improvement (ventilation measurement), four rejected outright (UV/light spatial correlation, fluence-binning resolution, injection source size, and now Phase 2 convergence quality). Every mechanism tested so far, spatial, geometric, or numerical, has failed to move Method B's answer toward Method A's by more than a rounding error, while Method B's own answer stays locked in a narrow band (62–70/hr) no matter what gets changed. That stability is itself the clue: the search has narrowed away from anything about the room or the run, and toward the two methods' underlying math — fitting a rate from a transient curve versus inferring one from a steady-state ratio — as the next place to look. Method A's own decay curves already show signs of not being a clean single exponential; whether that multi-mode behavior is large enough to account for the remaining gap is the open question the log picks up next.

— log continues at Entry 6